## Case Study: Ransomware Attack with Real-World Example

Ransomware is a cybercrime where hackers encrypt an organization's data, demanding money from the victim to decrypt the information. One real-world example is the WannaCry ransomware attack, which affected thousands of computers globally, including hospitals and business entities.

The process usually starts with an email or vulnerability exploit. In most instances, the victim is not aware that they have opened a phishing email or clicked on an exploit. The ransomware then starts to spread across the network, encrypting vital information. After encrypting the information, it sends a ransom note demanding money, usually in the form of cryptocurrency, in exchange for the decryption key.

The common targets for ransomware attacks are hospitals, small businesses, and learning institutions, which heavily rely on their data. These entities usually lack robust security. In the case of the WannaCry attack, hospitals were affected, with patient information becoming unavailable, causing appointment cancellations.

The victims usually pay the demanded money, which is usually in cryptocurrency, hoping to regain access to their information. In some instances, the hackers can shut down the entire system, causing significant losses. In the WannaCry attack, hospitals, small businesses, and learning institutions were affected, with patient information becoming unavailable.